

Security analysis of digital image watermarking using deep learning inspired LSB and chaotic S-Box in cyber security

Muhammad Zubair Shoukat ^a , Zhaofeng Su ^{a,*}, Jihad Ali ^b ,*

^a School of Computer Science and Technology, University of Science and Technology of China, No. 100 Fuxing Road, Shushan District, 230093, Hefei, Anhui, China

^b Department of AI Convergence Network, Ajou University, Suwon 16499, South Korea

ARTICLE INFO

Keywords:

Cryptography
Digital image watermarking
Holistically nested edge detection
Least significant bit

ABSTRACT

Information Technology (IT) has experienced remarkable advancements, this revolutionized the way of living, working and talking. One most momentous innovation is the social media communication and data sharing. The increasing prevalence of vast data sharing in the modern digital landscape has raised significant concerns regarding data privacy. Cryptography and image watermarking are essential components in the domain of security, specifically in providing services for multimedia protection. These technologies play crucial roles in opposing the integrity, confidentiality, and ownership of digital multimedia assets such as images, videos, and audio files. This research combines Least Significant Bit (LSB) and Holistically Nested Edge (HED) with chaos-based Substitution Box (S-Box) to enhance the strength of digital data security. The proposed technique summarizes the benefits of suitable positions by HED and high capacity and robustness against image processing attacks by LSB and encryption services by chaotic S-Box. The performance metric of proposed technique is calculated on Lena, Baboon and Pepper images. whereas to analyze the robustness Bit Error Ratio (BER) is calculated against different attacks i.e., speckle, gaussian, salt and pepper noise and cropping attacks on different images. The outcome shows that proposed technique opposing most of the attacks and legitimize it is effectiveness. The average value of Peak Signal to Noise Ratio (PSNR) and Structure Similarity Index (SSIM) are 53.15 dB and 0.9 which indicates invisibility of watermark signal in carrier image. The average value of Normalized Correlation (NC) is 0.9 which indicates proposed technique has strong robustness.

1. Introduction

Multimedia technology has been advancing rapidly, and various forms of media, including images, are becoming more prevalent in digital communication and entertainment. The extensive availability of IT tools and systems, such as computers, smartphones and high-speed internet, have become increasingly affordable and widely available. The extensive utilization of social networking platforms and the exchange of diverse multimedia have been instrumental in driving this shift. Besides considerable benefits brought by technological progress and the rise of online commerce, there are corresponding worries regarding security and privacy. Various challenges and risks are opened while sharing content on social media platforms [1]. Due to extensively usage of cameras and smart phones to capture moments and shared with community on multiple social media platforms, subsequently broadcasting of digital content attract intruders to transpire it. Now a days, protection of digital content become challenge for both the academia and industry due to several evolving factors in the digital landscape. With the rapid

growth of the internet and advancements in technology such as AI and machine learning, the methods for accessing, distributing, and manipulating digital content have become more sophisticated. As a result, safeguarding digital assets, intellectual property, and sensitive information is an increasingly complex task.

To address this issue there is a field of security named as cryptography, by which we can secure content through hidden codes of transmission. Cryptography is the development of cryptosystems that provide security services such as secrecy, integrity, and authentication. Furthermore, steganography and digital image watermarking are sub-fields of cryptography that provide digital content secrecy by encoding ownership in digital content. During data encoding, digital data is transformed into a format that intruders cannot understand. To protect digital content, watermarking is flatterring a progressively important means for securing digital content. Besides the conventional purposes of watermarking, researchers are exploring its application in various

* Corresponding authors.

E-mail addresses: zfsu@ustc.edu.cn (Z. Su), jehadali@ajou.ac.kr (J. Ali).

<https://doi.org/10.1016/j.jisa.2025.104209>

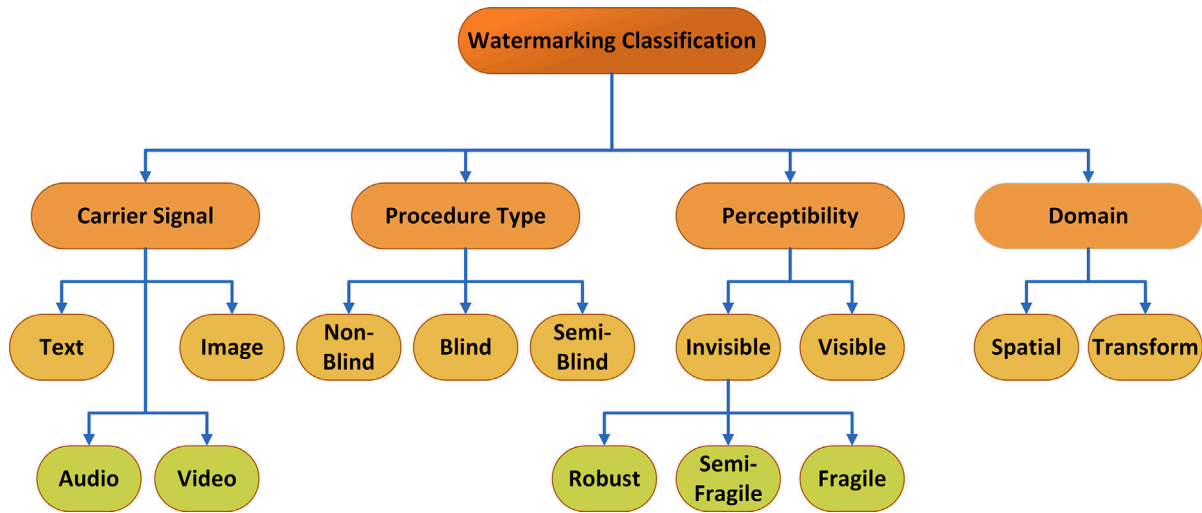


Fig. 1. Image watermarking taxonomy.

cybersecurity areas such as intrusion detection and data accuracy. Digital image watermarking falls within the realm of information security, ensuring data integrity by implanting a watermark signal within the original image, subsequently extracted to authenticate ownership.

Various types of watermark signals exist, tailored to different watermarking methods. These methods include visible watermarking, which safeguards intellectual property, and invisible watermarking, concealing the signal to confirm ownership and validate content. Fig. 1 illustrates the categorization of watermarking.

Image watermarking predominantly operates in two domains: spatial and frequency. The spatial domain is simpler, robust and have higher capacity, contrasts with the frequency domain, which is more intricate, robust, but with less capacity. The techniques in spatial domain image watermarking include the LSB, known for its simplicity and high embedding capacity [2]. In the frequency domain, techniques like Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) are prominent. DCT, although complex, can lead to loss in the original image during reconstruction. On the other hand, DWT involves dividing the image into four bands, presenting high complexity and functioning as a lossless compression technique.

The robustness of digital image watermarking lies in the ability of the extracted watermark signal to remain intact. Robust watermarking withstands external attacks, whereas fragile watermarking is susceptible to disturbances during the attacks. Based on embedding and extraction process watermarking is categorized as blind watermarking does not need any additional information to retrieve the watermark from watermarked image. Semi-blind watermarking needs partial information to retrieve the watermark from watermarked image. Non-blind watermarking, is the conventional approach, it requires entire awareness of both the original and the watermark image to retrieve the watermark signal.

Watermarking process constitute of two processes i.e., watermark embedding and watermark extraction as shown in Fig. 2. During watermark embedding extra information is insert into original media, the quality of the media degraded. In order to preserve the quality of the media, the watermark should be inserted in such a way the quality of the media preserved. The parameters to balance the quality of the media is PSNR and SSIM. The quality of an embedded watermark is seemed as original to the carrier medium [3,4]. Currently, researchers are finding the areas to insert watermark in such way that it could not distort the quality of original image. While in watermark extraction, the watermark signal is safely recovered without any distortion is the main concern of the watermarking technique. The factor on which quality of watermark depends is NC.

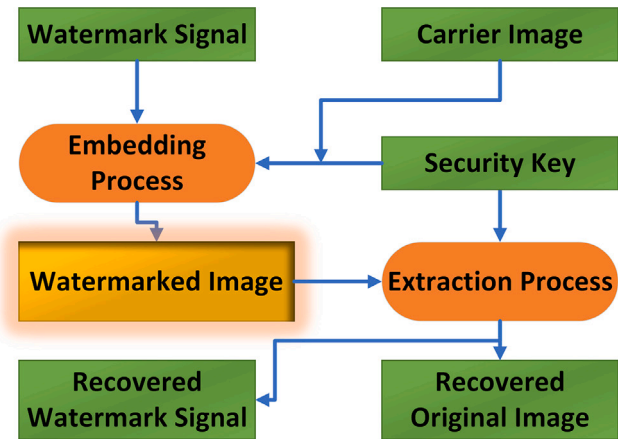


Fig. 2. Generic diagram of watermarking.

In this research, a deep learning inspired image watermarking approach is designed to secure the digital content. HED is used to find the sharp changes in the media. After applying convolutional layers to the image, HED separates it into five side output layers, allowing for the detection of sharp discontinuities. There is a scale difference between the two sides' edge predictions, lower layers pick up on finer features, while higher layers cover more ground. Finally, the output with abrupt discontinuities is obtained by combining all of the side output layers. LSB is used to embed watermark signal at sharp changing positions in original image, and it has high payload capacity to embed watermark. Moreover, a chaotic S-Box is used to enhance the security of the watermark signal.

The major contribution of the research is:

- This research presented deep learning-inspired technique to find the sharp discontinuities which helps to find best part of the image where watermark can be added maximum data in carrier image.
- The purpose of this finding is to embed watermark signal in that area where it is impossible to detect easily. This makes it hard to find and increase the watermark robustness and make minimum effect on image visualization. LSB is used due to low computational efficiency during embedding and extraction process. The security of the watermark is enhanced by using the S-Box to encrypt the watermark signal before embedding.

The remaining document is structured as outlined below: Section 2 presents the existing literature, while Section 3 covers the foundational studies referenced in this manuscript. Section 4 describe in depth the proposed technique, followed by Section 5, which outlines the results and in-depth discussion of the proposed scheme. Finally, Section 6 describe the paper's conclusion.

2. Related work

Keeping in mind the different factors i.e., image quality, robustness, security and computational efficiency of numerous image watermarking techniques have been introduced having ironic properties of PSNR, imperceptibility and robustness. Imperceptibility defines the quality of an embedded watermark being undetectable or invisible to human perception within the carrier medium [3,4]. Faheem et al. [5] proposed technique improves security and robustness by using campus edge detector to detect the edges of cover image and LSB mechanism to embed watermark image. Moreover, to improve the security of the during watermarking chaotic encryption is used.

LSB algorithm and AuSR2 scheme used in [6]. AuSR2 is used for extra security on multiple layers and recovers each pixel independently without any improvisation. The outcomes are very effective in terms of self-recovery and image authentication. The only issue is that PSNR value is not satisfactory which leads to low visual quality of watermarked image. DWT and matrix decomposition-based method is explored by wang et al. [7]. To enhance the security multi-level DWT and Chaotic maps are used. Proposed algorithm shows good performance in terms of resisting attacks, besides this the major drawback of the technique is that it consumes more power and computationally expensive because researcher joined different optimizing algorithm to design this watermarking technique.

Rishi et al. [8] used a machine learning based technique to embed watermark in which position of watermark embedding is selected by a random number generator named as Mersenne Twister Generator which reduce the computational complexity. The outcomes achieve robustness and but the technique is compromised with imperceptibility and generates low quality carrier image due to which watermark is easy to detect. Although, it shows good results against different attacks. Mohammad et al. [9] proposed watermarking method in transform domain by using canny edge detector. In this method watermark is implanted in the diagonal of DCT after converting in encrypted vector and insert into edge vector. Diagonal implantation is used to avoid temptation but the outcome produces low quality which leads to image processing assaults.

CNN based approach is used to detect edges in [10], in which the cover image is pre-processed by machine learning based approach by masking the last five bits to every pixel. After that, prominent parts are obtained by using the CNN, LSB is used to add watermark into host image by adding more bits into the sharp edges and less bits into less sharp edge of cover image. Due to the high payload this technique compromised with Visual quality of the image.

Table 1 shows comparison of different technique used previously. The comparison is carried out in terms of following factors watermarking technique, Advantages, Disadvantages, Imperceptibility and security.

2.1. Current challenges

Researchers have made great contribution in recent years in overcoming the challenges of multimedia content authentication. But there is more space available in few potential challenges.

- Robustness, capacity and imperceptibility remains on top for researchers.
- Cryptography plays a crucial role in watermark insertion for enhancing the security.

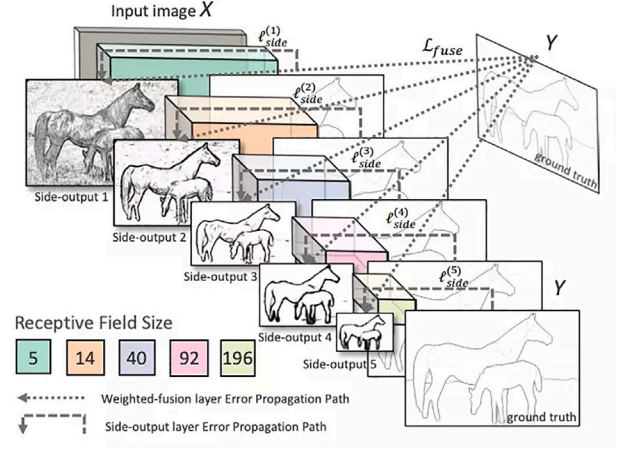


Fig. 3. HED network architecture [22].

- Trigger immediate alerts or actions upon detecting an attack.
- Finding efficient method is essential because embedding and extraction process can be computationally expensive.
- Requiring new methods with strong generalization capabilities.

3. Theoretical background

3.1. Holistically Nested Edge Detection (HED)

Robert, Prewitt and Sobel are most commonly pioneer methods in early edge detection, while the extensively utilized canny edge detector excels in identifying the sharp changing in image. To find the sharp discontinuities of image in X-direction and Y-direction Prewitt and Sobel Edge detector are used. Here, we use HED to find sharp discontinuity in image, which is a deep learning model based on multi-layer convolutional network.

- HED makes the side outputs of the convolutional layer.
- The final prediction is generated by the combination of each side output layer made by HED.
- Because of feature maps are produced at each layer have varying sizes, the model examining the image at different scales.

Fig. 3 describes in detail the utilization of HED based multi-layered CNN to generate the final edge-to-edge output image. Where X and Y are the input and output images respectively [22].

Instead of using X and Y direction to detect discontinuities, HED works in such a way that the foreground and background features can be distinguished from one another. HED discover edges define pixels and allocate an associative weight with respect to the sum of pixels that are found in image. Eqs. (1)–(4) given below express how HED works.

$$L_{side}(W, w) = \sum_{q=1}^Q \beta_q l_{side}^{(q)}(W, w^{(q)}) \quad (1)$$

Eq. (1) shows the function to collect all network layer parameters where w and W denotes every side output layer's classifier and corresponding weights respectively. L_{side} is objective function, Q is total number of layers and q is for individual output layer. β_q and $l_{side}^{(q)}$ denotes associative learning rate and lose level function respectively. L_{fuse} computes the weights associated with each layer L_{side} .

$$L_{fuse}(W, w, h) = Dist \left(Y, \hat{Y}_{fuse} = \text{sigmond} \left(\sum_{q=1}^Q h_q A_{side}^{(q)} \right) \right) \quad (2)$$

Table 1
Comparison of literature watermarking techniques.

Reference paper	Method	Advantages	Disadvantages	Imperceptibility (PSNR)	Security
[11]	LSB Hash Algorithm	High capacity, Data extraction effectively from watermark	Low robustness	–	Low
[12]	LSB	Watermarked image quality is high, Fast speed	Only B component used for embedding	47.6 dB	Above average
[13]	2DWT+DCT	Low bit error rate	Less BCR Value	40 dB	High
[14]	LSB+DWT+SVD	Capacity to restore altered areas	Less robust against geometric attacks	48.13 dB	Low
[15]	LSB+DWT+DCT	High visual quality	Less robust against geometric attacks, No temper detection	41 dB	High
[16]	ISB+Machine Learning	Good robustness based on NCC	Less robust against geometric attacks	30 DB	Average
[17]	DWT+SVD+SVN+Machine Learning	Provides 2nd transmission medium over LTE network	High computational complexity	52.74 dB	Above average
[18]	DNN	High quality watermark image	Low embedding strength	41.07 dB	High
[19]	LSB+DNN+LZW,	High Payload, Visual imperceptible	Less robust against geometric attacks	40.27 dB	High
[20]	DNN+SS-GWO+ADWT	High payload	Resolution of reconstruct image is low, high Computational Complexity	19.80 dB	Low
[21]	DNN	High capacity.	Maximize the Key sample will minimize the performance of DNN	43.10 dB	High

In Eq. (2) $\hat{Y}_{fuse}$ defines edge map prediction computed by CNN. The sigmoid activation function helps in this by squashing the output values to the range [0, 1], allowing the network to interpret these values as probabilities. h_q fusion weight and $A_{side}^{(q)}$ activation function of side output layer is calculated over Sigmond function. 0 and 1 are used as probability for edge classification. Both (1) and (2) are used to minimize the objective function using stochastic gradient descent.

$$(W, w, h)^* = \underset{[L_{side}(W, w), L_{fuse}(W, w, h)]}{argmin} (L_{side}(W, w) + L_{fuse}(W, w, h)) \quad (3)$$

$$\hat{Y}_{HED} = \sum_{q=1}^Q (\hat{Y}_{side}^{(q)}, \hat{Y}_{fuse}^{(1)}, \hat{Y}_{fuse}^{(q)}) \quad (4)$$

Testing results are computed using updated weights $(W, w, h)^*$ where outcomes from the CNN results are calculated for each fusion $\hat{Y}_{fuse}^{(1)}$ and side $\hat{Y}_{fuse}^{(q)}$ layer.

3.2. Least significant bit

LSB is indeed the best and most commonly used method in time domain watermarking. Its popularity stems due to its simplicity and ease of implementation. LSB is based on pixel level computation and two or three bits are used to embed watermark [23]. For embedding and extraction of watermark XOR operation is performed on carrier image. While LSB-based methods are straightforward and offer high embedding capacity, they are more susceptible to attacks compared to more complex techniques. Since the least significant bit holds the least weight in determining pixel color or intensity, modifying it has a subtle impact on the visual appearance of the image [24]. LSB offers high payload capacity which makes it robust, which means that it can accommodate large amount of data in carrier image. Fig. 4 describes the pixel value of LSB.

Fig. 5 shows the actual operation executed in LSB. Right shift operation is performed on four least significant bits of watermark image and zero is embedded on four least significant bits of carrier image. Final image is retrieved after combining the LSB and MSB bits of both the carrier and watermark image.

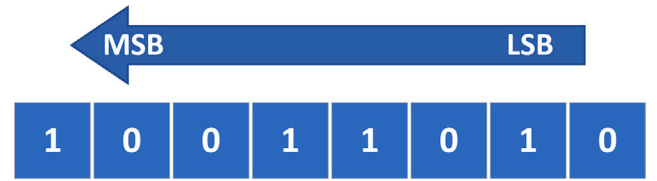


Fig. 4. LSB value.

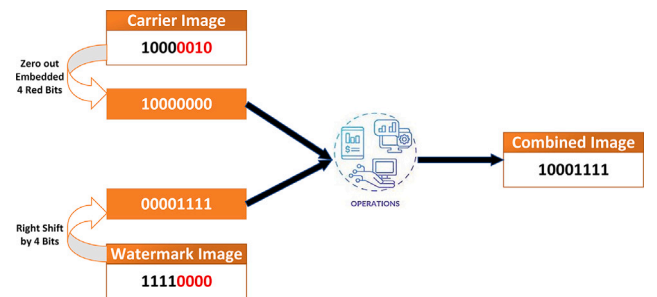


Fig. 5. Block diagram (LSB embedding process).

3.3. Chaotic substitution box

S-box plays fundamental role in cryptosystems and creates misconception. Indeed, chaos theory has found applications in modern cryptosystems due to its inherent characteristics of nonlinearity and sensitivity to initial conditions. Random number creation is the main application of chaotic maps which leads to further operations on output numbers. Depending on the initial condition random and different numbers can be generated. Now a days these mechanisms are typically used in S-box design due to their diverse nature. Favorable outcomes can be obtained from chaos-based S-box, these systems are also computationally affordable. The main purpose to use S-box is for encryption process in watermarking. Z. Bin Faheem et al. [25] proposed

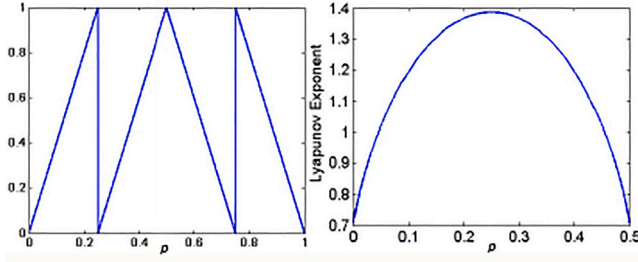


Fig. 6. PWLCM plot [27].

S-box based on piecewise linear chaotic map (PWLCM). In proposed approach, PWLCM based S-box is used to scramble the bits of the watermark signal. PWLCM has broader random behavior and constant density function [26]. Equation and ranges are given below.

$$x_{n+1} = \begin{cases} \frac{x_n}{p}, & 0 \leq x_n < p \\ \frac{x_n - p}{0.5 - p}, & p \leq x_n < 0.5 \\ \frac{1 - p - x_n}{0.5 - p}, & 0.5 \leq x_n < 1 - p \\ \frac{1 - x_n}{p}, & 1 - p \leq x_n < 1.0 \end{cases} \quad (5)$$

where $X_0 \in [0, 1)$ and p is the control factor $p \in (0, 0.5)$. PWLCM plot is shown in Fig. 6.

4. Proposed HED based watermarking technique

Two steps are involved in watermarking one is embedding of watermark signal in carrier image and the other is extraction of watermark signal from the carrier image.

In our proposed method HED is used to detect the sharp curves of cover image in which watermark is to be added and LSB technique is used to embed watermark into cover image. In this approach mathematical operations are less during embedding and extraction due to the low calculation complexity of LSB. The whole process carries out in two step one is embedding and the other one is extraction discussed below. HED finds the most appropriate places where watermark can be added while LSB is used for watermarking insertion and extraction as shown in Fig. 7. Performance and resilience are tested on many photos using various techniques, including geometrical and image processing assaults. It has been discovered that the suggested method is resilient to various visual assaults. The pseudo-code for the proposed technique is given in Algorithm 1 and 2.

The embedding and extraction process for the proposed watermarking scheme are as follow.

4.1. Embedding

1. Image of 512×512 and 32×32 are used as original image and watermark image respectively.
2. Dividing the image into 16×16 size blocks.

- Image Partitioning:

- The carrier image of 512×512 is divided into 32×32 blocks, each block contains 16×16 pixels. This is done by partitioning of original image into a grid-like structure. where each block defines a portion of the original image.

- Extracting Blocks from image:

Algorithm 1 Embedding Process

1. Divide image into 16×16 blocks


```
def div(img, blk_size):
    blocks = []
    height, width = img.shape[:2]
    for y in range(0, height, blk_size):
        for x in range(0, width, blk_size):
            blk = img[y:y+blk_size, x:x+blk_size]
            blocks.append(blk)
    return blocks
```
2. Edge Detection with HED


```
def hed(img):
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    edges = hed.detectEdges(gray)
    return edges
```
3. Embed watermark bits based on edge capacity


```
def embed_wm(blocks, wm, cap_threshold):
    wm_blks = []
    for blk in blocks:
        cap = calculate_cap(blk)
        if cap >= cap_threshold:
            wm_blk = embed_less_bits(blk, wm)
        else:
            wm_blk = embed_more_bits(blk, wm)
        wm_blks.append(wm_blk)
    return wm_blks
```
4. Shuffle watermark bits using S-box


```
def sb_shuffle(wm):
    shuffled_wm = wm # placeholder
    return shuffled_wm
```
5. Multiply 16×16 blocks with 32×32 blocks of watermark


```
def mul_blk(orig_blks, wm_blks):
    multiplied_blks = []
    for orig_blk, wm_blk in zip(orig_blks, wm_blks):
        multiplied_blk = np.multiply(orig_blk, wm_blk)
        multiplied_blks.append(multiplied_blk)
    return multiplied_blks
```
6. Combine LSB and MSB


```
def comb_lsb_msb(multiplied_blks):
    comb_blks = []
    for blk in multiplied_blks:
        comb_blks.append(blk)
    return comb_blks
```
7. Final watermarked image


```
def construct_wm_img(original_image, comb_blks):
    height, width = original_image.shape[:2]
    wm_img = np.zeros((height, width), dtype=np.uint8)
    index = 0
    for y in range(0, height, 16):
        for x in range(0, width, 16):
            wm_img[y:y+16, x:x+16] = comb_blks[index]
            index += 1
    return wm_img
```

- Based on the size of block 16×16 calculating starting indices of rows and columns for each block. Each block $B_{i,j}$ where i and j are rows and columns can mathematically be represented as.

$$B_{i,j} = \text{Image}[16 \times i : 16 \times (i + 1), 16 \times j : 16 \times (j + 1)]$$

- Each $B_{i,j}$ is 16×16 sub image of carrier image.

3. Insertion of side output layer after convolutional layer to find the discontinuities.

$$L_{side}(W, w) = \sum_{m=1}^M \alpha_m l_{side}^{(m)}(W, w^{(m)})$$

Image level loss function denoted by l_{side} for side outputs. Each side output layer is also associated with classifier, in which

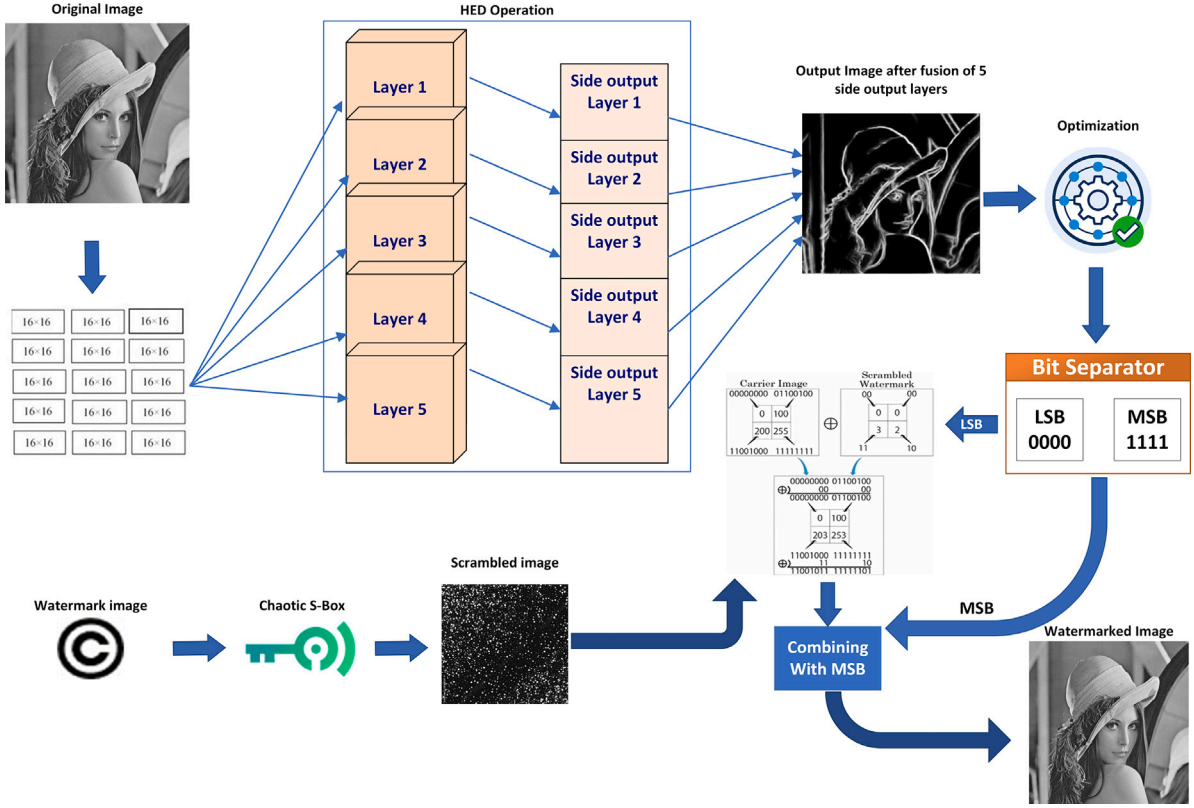


Fig. 7. Proposed watermark technique.

weights are represented as $W = (W^{(1)}, w^{(m)})$. These side outputs correspond to edge predictions at different scales. Lower layers tend to capture finer details (smaller edges), while higher layers capture more global and abstract edge information.

4. Applying deep supervision at each side-output layer. Each layer applies convolutional filters, capturing patterns and edges. $I(x, y) * K(u, v) = O(x + u, y + v)$ is the convolutional operation.
5. Each side outputs are fused together to combines the information from different layers using weighted-fusion layer, resulting in more detailed sharp edges.

$$L_{fuse}(W, w, h) = Dist(Y, \hat{Y}_{fuse})$$

$$\hat{Y}_{fuse} \equiv \sigma \left(\sum_{m=1}^M h_m \hat{A}_{side}^{(m)} \right)$$

where $h = (h_1, \dots, h_m)$ is fusion weight and $Dist(.,.)$ represents the distance between ground truth and fused prediction.

6. After optimization bits are separated into LSB and MSB, meanwhile best portion of image is found where watermark bits can be added. Some portions have ability to carry more bits rather than others.

- Less bits are embedded in portion having less ability to carry watermark bits $W \geq \max(W/2)$
- More bits are embedded in portion having more ability to carry watermark bits $W < \max(W/2)$

7. To increase the security 16×16 S-box is used to shuffle the bits of watermark signal of size 32×32 .

- Initialization:

- Denotes watermark signal as $W_{32 \times 32}$.

- Generate shuffled watermark signal containing all zero values denotes as $W_{shuffled}$

• Bit Mapping using S-Box:

- Bit wise iteration over original watermark signal $W_{32 \times 32}$.
- In 32×32 grid determine column and row for each bits i for rows and j for column.
- To determine the new position of each bit in S-Box corresponding entry is denoted as $S[i][j]$.
- Update the corresponding position in shuffled watermark signal $W_{shuffled}$ with the value of the current bit.
- Repeat the above step for all bits in watermark signal.
- $W_{shuffled}$ now contains the bits rearranged according to the entries in S-box.

Mapping process can mathematically expressed as:

$$W_{shuffled}[S[i][j]] = W_{32 \times 32}[i][j]$$

8. Each 16×16 block of original image is multiplied with 32×32 block of watermark image.
9. To construct final watermark image LSB and MSB are combined.

4.2. Extraction

1. Watermarked Image of 512×512 is loaded in HED. HED consists of multiple layers, binary images with highlighted discontinues are obtained after application of HED.

$$HED(\text{watermarked Image}) = \text{Edge Image}$$

Algorithm 2 Extraction Process

1. Divide Watermarked image into 16 $\times$ 16 blocks


```
def div(wm_img, wm_blk_size):
    wm_blk = []
    height, width = wm_img.shape[:2]
    for y in range(0, height, wm_blk_size):
        for x in range(0, width, wm_blk_size):
            wm_blk = wm_img[y:y + wm_blk_size, x:x+wm_blk_size]
            wm_blks.append(wm_blk)
    return wm_blks
```
2. Edge Detection with HED


```
def hed(wm_img):
    gray = cv2.cvtColor(wm_img, cv2.COLOR_BGR2GRAY)
    edges = hed.detectEdges(gray)
    return edges
```
3. Extract LSB and MSB of watermarked image


```
def ext_lsb_msb(wm_img):
    lsb = np.mod(wm_img, 2)
    msb = wm_img - lsb
    return lsb, msb
```
4. Divide original image into 16x16 blocks

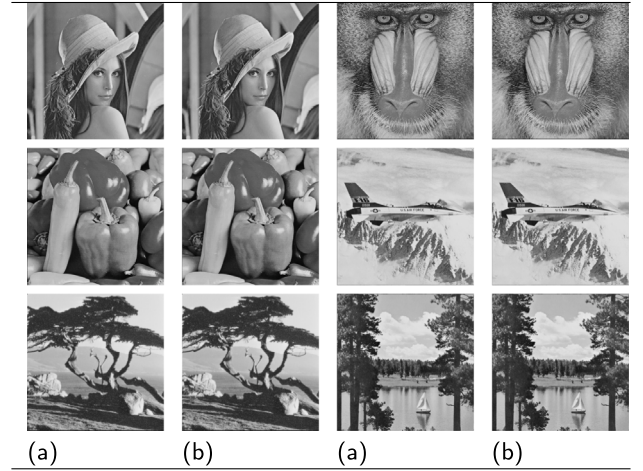

```
def div(img, blk_size):
    blocks = []
    height, width = img.shape[:2]
    for y in range(0, height, blk_size):
        for x in range(0, width, blk_size):
            blk = img[y:y+blk_size, x:x+blk_size]
            blocks.append(blk)
    return blocks
```
5. XOR operation on LSB of original and watermarked images


```
def xor(ori_lsb, wm_lsb):
    xor_result = cv2.bitwise_xor(ori_lsb, wm_lsb)
    return xor_result
```
6. Decryption process using S-box


```
def s_box_decrypt(xor_result):
    decrypt_wm = xor_result
    return decrypt_wm
```

Table 2

Comparison of original and watermarked images (Original Images (a) Watermark Images (b)).



a deep learning-inspired approach and LSB. After inspecting several properties, it is found that the visual appearance of host image and the resistance against different attacks are the most significant. Furthermore, hardware plays an important role while concluding results. For proposed technique experiments are conducted on the machine with Ram 8 GB RAM, core-i5 (9th gen) and PyCharm Community addition is used to perform the operations.

Multiple carrier images are used to conduct experiments and find the effectiveness, for this carrier images are taken from <https://engineering.nyu.edu/> (accessed on 15 Dec 2023) and the size of the carrier images and watermark image are 512×512 and 32×32 respectively. Comparison of original and watermarked images can be seen in Table 2.

5.1. Performance evaluation measure

5.1.1. Visual appearance measure

Visual quality defines the visual appearance to measure the Visual quality of both carrier and watermarked image PSNR and SSIM is used.

Peak Signal to Noise Ratio (PSNR): Higher the PSNR value represents higher quality of image. PSNR value approaches to infinity when Mean Square Error (MSE) approaches to Zero [28]. The minimum value of PSNR is 0 and theoretically has no maximum value but when we talk about experimental work it is limited by the precision of image pixels. Higher the PSNR value represents higher quality of image. And it is calculated in (db) Mohammad et al. [9] shows the calculation of PSNR.

$$PSNR = 10 \log_{10} \left(\frac{255^2}{MSE} \right) \quad (6)$$

In Eq. (6) 255 represents the highest grayscale value while MSE is mean square error. PSNR value approaches to infinity when Mean Square Error (MSE) approaches to Zero.

$$MSE = \frac{1}{M \times N} \sum_{m=1}^M \sum_{n=1}^N e(m, n)^2 \quad (7)$$

The calculation of PSNR and comparison with other techniques given in Table 3 and graphical comparison in Fig. 8. In [8] technique is used to embed the watermark based on random number generator called Mersenne Twister Generator. This technique reduces the complexity but generates resultant image having low quality with PSNR value 40.1 db, 39.7 db for lena and baboon images respectively. On the other hand, in [10] CNN based approach is used by masking the

2. Pixel separation of watermarked image into LSB and MSB. P is the pixel value then 8-bit binary number can be represented as:

$$P = Y_7 Y_6 Y_5 Y_4 Y_3 Y_2 Y_1 Y_0$$

where

$$LSB = Y_0$$

$$MSB = Y_7 Y_6 Y_5 Y_4 Y_3 Y_2 Y_1$$

3. XOR operation is carried out on the LSB of corresponding pixels of both watermarked and original image. Lets assume C_{LSB} for carrier image LSB and W_{LSB} for watermarked image LSB. The XOR operation will be:

$$XOR = W_{LSB} \oplus C_{LSB}$$

4. Decryption process is carried out and rearranged the scrambled bits of watermark image using S-Box. XOR result obtained in previous step undergoes decryption process as:

$$W_{scrambled} = Decryption(W_{watermark})$$

$$W_{reconstructed} = Scramble(W_{scrambled})$$

5. Watermark is retrieved and confirmed that the host image is in its original condition. If the visual quality of host image is disturbed it means host image is unsafe and attacked by intruders.

5. Experimental results and discussion

Following section presents in depth discussion of outcomes after comparing the results of our proposed image watermarking using HED

Table 3
Visual appearance measure PSNR.

Technique	Lena image	Baboon image
[5]	55.4	52.7
[8]	40.1	39.7
[10]	43.5	44.2
Proposed	54	52.3

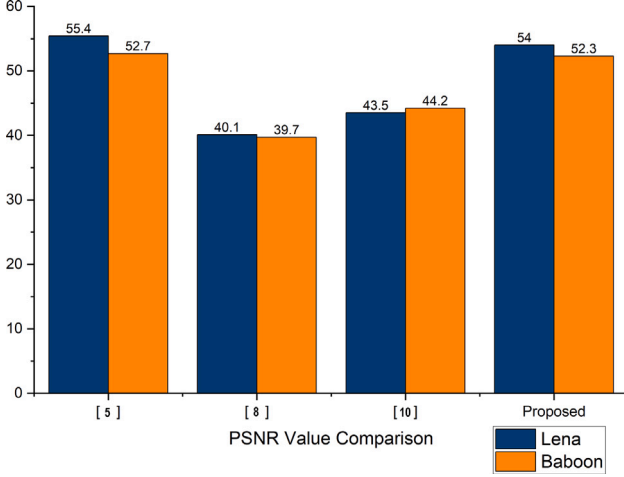


Fig. 8. Graphical comparison of PSNR for Lena and Baboon image.

last five bits in each pixel while LSB is used to add watermark. As, more bits are added in prominent part of the image which decrease the visual quality of the image and obtain maximum 43.5 db for lena and 44.2 db for baboon image. By comparing both of these method with proposed one the PSNR value for lena image is 54 and 52.3 for baboon and almost near to [5] which shows that the proposed method has good preservation of image quality.

Structure Similarity Index Measure: SSIM is used to find the similarity between two images and is interrelated to the excellence of human visual system (HVS). SSIM is designed on the basis of following three factors loss of correlation, luminance distortion and contrast distortion. Keeping in view the distortion of any image [29].

$$SSIM(X, Y) = [I(X, Y) \cdot C(X, Y) \cdot S(X, Y)] \quad (8)$$

where $I(X, Y)$ is luminance distortion, $C(X, Y)$ is contrast distortion and $S(X, Y)$ structure correlation.

$$I(X, Y) = \frac{2 \cdot \mu_x \cdot \mu_y + C_1}{\mu_x^2 \cdot \mu_y^2 + C_1} \quad (9)$$

X and Y are images and their average luminance values defined by μ_x and μ_y where C_1 is constant. μ_x and μ_y can be calculated as follows.

$$\mu_x = \frac{1}{T} \sum_{i=1}^T x_i, \mu_y = \frac{1}{T} \sum_{i=1}^T y_i \quad (10)$$

The term is a function that assesses the similarity in brightness between the two images μ_x and μ_y . The maximum value can be 1 only if $\mu_x = \mu_y$.

$$C(X, Y) = \frac{2 \cdot \sigma_x \cdot \sigma_y + C_2}{\sigma_x^2 \cdot \sigma_y^2 + C_2} \quad (11)$$

$$\sigma_x = \frac{1}{T-1} \sum_{i=1}^T (x_i - \bar{x})^2, \sigma_y = \frac{1}{T-1} \sum_{i=1}^T (y_i - \bar{y})^2 \quad (12)$$

σ_x and σ_y are standard deviations of pixel intensities in X and Y and its maximum value can be 1 only if $\sigma_x = \sigma_y$. Where C_2 is constant

Table 4
Visual appearance measure SSIM.

Technique	Lena image	Baboon image
[5]	0.99	0.99
[8]	0.99	0.99
[10]	0.97	0.98
Proposed	1	0.99

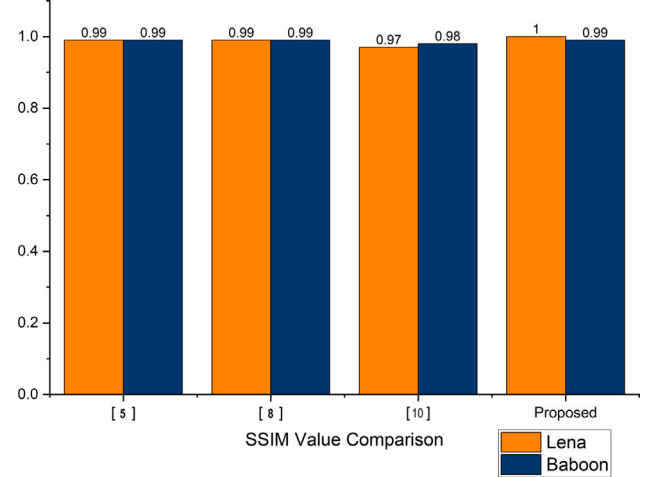


Fig. 9. Graphical comparison of SSIM for Lena and Baboon image.

in Eq. (13).

$$S(X, Y) = \frac{\sigma_{xy} + C_3}{\sigma_x \cdot \sigma_y + C_3} \quad (13)$$

$$\sigma_{xy} = \frac{1}{T-1} \sum_{i=1}^T (x_i - \bar{x})(y_i - \bar{y}) \quad (14)$$

σ_{xy} correlation between X and Y and is another constant.

SSIM index values lies between [0,1]. If the value approaches to 0 defines no correlation between images, similarly if the values approach to 1 defines that the images are highly interrelated. Detailed calculation of SSIM given in [30].

Table 4 shows the comparison of SSIM values with existing techniques and graphical comparison in Fig. 9. SSIM values approaches to 1 for lena image which shows that the original and watermarked image are same. Furthermore, the value is higher as compared to existing literature i.e., for lena image proposed method obtained 1 while other methods maximum value is 0.99. For baboon image vale is 0.99 which is higher than [10] and equals to [5,8]. The overall results shows that the original and processed image structure, luminance distortion and contrast are well-retained and the quality of water mark image is very good and visually indistinguishable.

5.1.2. Robustness

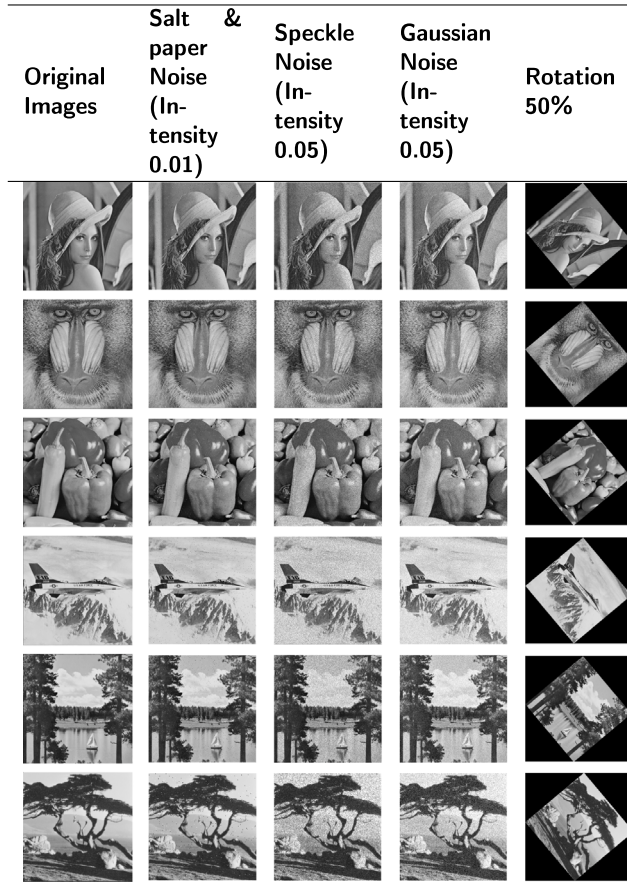
Robustness is often associated the resilience of a system against disparities in inputs and environmental factors. While communicating over cyberspace intruders can interrupt the multimedia content. It is very difficult to attain safety of our content on different social media platforms as well as over the internet. Robustness is measured by Normalized Correlation NC.

Normalize Correlation (NC): The value of NC depicts the safety of watermark carrier image. Image is more secure and recovered successfully as the value approaches to 1 or robustness of the watermarking image increase with the value of NC closed to one [3,31].

$$NC = \frac{\sum (X_i - \bar{X}) \cdot (Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2}} \quad (15)$$

Table 5

Before and after effects of images after applying different attacks.



Eq. (15) Shows normalized correlation equation in which X_i and Y_i are the intensities while $\bar{X}$ and $\bar{Y}$ are the mean average pixel intensities of X and Y images. Equations to calculate the mean average pixel intensities are given below.

$$\bar{X} = \frac{1}{N} \sum X_i \quad (16)$$

$$\bar{Y} = \frac{1}{N} \sum Y_i \quad (17)$$

In (16), (17) N shows the total number of pixels. Various attacks including image processing and geometrical attacks are applied to analyze the robustness of the proposed method. These attacks effect the visual appearance and the dimension of the images. Different images are used to test the robustness and multiple attacks are executed, to test and verify the effectiveness and it is observed that the proposed method is invulnerable to numerous attacks. In Table 5 results of before and after attacks of different images can be seen.

Deep learning and LSB based approach is proposed by Sinhal et al. in [19]. To select regions IWT transform and random block selection is used while LSB is used to embed the watermark. Furthermore, to increase security SHA-256 algorithm-based hash keys are applied on ROI and RONI. Outcomes are good for JPEG attacks rather than Median and S&P attacks. Maximum value of NC for median attack is 0.93 as compared to our proposed method which is 0.992. While for S&P attack NC value is 0.98 which is less than proposed technique.

To increase the imperceptibility and robustness two combined techniques are used in [32]. Initially DWT-DCT-SVD hybrid domain is used from embedding secondly for optimization multi-dimensional PSO algorithm is used. Furthermore, to increase the security chaotic map is adopted. The outcomes shows that the maximum NC values against

Table 6

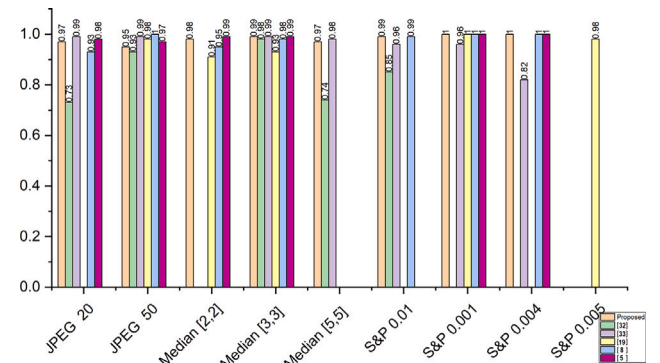
NC values comparison on image Lena.

Attack	Strength	Proposed	[32]	[33]	[19]	[8]	[5]
JPEG	20	0.97	0.73	0.99	–	0.93	0.98
JPEG	50	0.95	0.93	0.99	0.98	1	0.97
Median	[2,2]	0.98	–	–	0.91	0.95	0.99
Median	[3,3]	0.99	0.98	0.99	0.93	0.98	0.99
Median	[5,5]	0.97	0.74	0.98	–	–	–
S&P	0.01	0.99	0.85	0.96	–	0.99	–
S&P	0.001	1	–	0.96	1	1	1
S&P	0.004	1	–	0.82	–	1	1
S&P	0.005	–	–	–	0.98	–	–

Table 7

Lena image NC against attacks.

Attack	Strength	Proposed	[34]	[5]
Gaussian	0.005	0.98	0.97	0.99
Gaussian	0.1	0.90	0.37	0.97
Speckle	0.001	1	1	1
Speckle	0.01	0.99	0.99	0.99
Rotation	50	1	1	1
Cropping	25%	0.99	0.95	1
S&P	0.001	1	1	1
S&P	0.01	0.99	0.99	0.99
Median filter	[3, 3]	0.99	1	0.99
Median filter	[5, 5]	0.97	0.92	0.98

**Fig. 10.** Graphical comparison of NC against different attacks on image Lena.

JPEG, Median and S&P attacks approach to 0.93, 0.98 and 0.85 respectively which are less than our proposed method. Mostly SVD based watermarking techniques are not much resistant against attacks keeping in view another SVD based technique proposed by [33] in which extra steps are not included that make it robust against different attacks. This technique follows sharp frequency localized contourlet transform (SFLCT) for both watermark and host images and got significant results against JPEG attacks as compared with our HED based technique.

While performing JPEG compression at intensity level 20 the proposed technique has NC values 0.97 which slightly decrease to 0.95 with increase in intensity level to 50. Comparing with [8,32] the proposed method has higher NC while for others it is just comparable. On the other hand, proposed method upholds the maximum similarity NC value of 1 for 0.001 and 0.005 intensity during salt and pepper attack which is greater than [32,33] and similar to other literature results for some points. For median filtering value slightly increases from 0.98 to 0.99 with increase the size of filter form [2,2] to [5,5]. Comparing with existing work the results are higher in case of S&P attack. Overall, the results of NC value shows that the method uploading the maximum level of recumbence with original image. In Table 6 several existing gray scale image watermarking techniques are compared with proposed technique which shows the NC values comparison.

Table 8
Pepper image NC against attacks.

Attack	Strength	Proposed	[5]	[35]
Speckle	0.05	0.98	0.98	0.99
Gaussian	0.01	0.96	0.98	0.99
S&P	0.05	0.98	0.98	0.99
Median Filter	[3, 3]	0.99	0.99	0.99

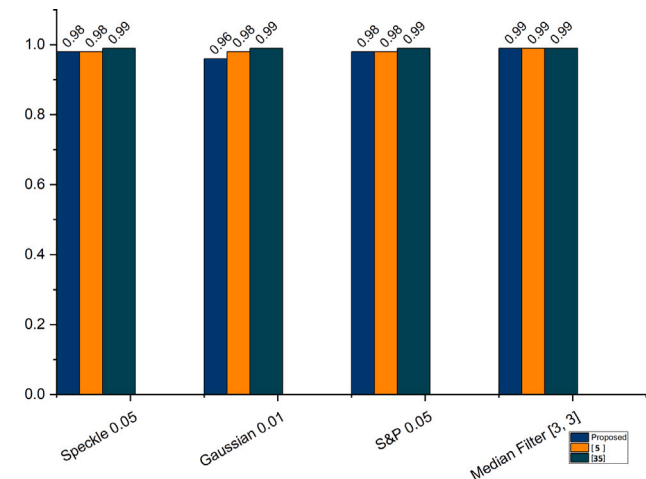


Fig. 11. Graphical comparison of NC for various attacks on the Pepper image.

Table 9
Baboon image NC against attacks.

Attack	Strength	Proposed	[5]	[35]
Speckle	0.05	0.98	0.98	0.99
Gaussian	0.01	0.97	0.98	0.99
S&P	0.05	0.98	0.98	0.99
Median Filter	[3, 3]	0.99	0.99	0.94

In Table 7 results of different attacks on lena image are compared with existing literature i.e., while median filtering attack with the kernel size [3, 3] the achieved value is 0.99 which shows the strong similarity with original image but increase in kernel value slightly decrease the similarity. Gaussian attack shows the slightly decrease in values as intensities increased which is much better as compared to [34]. S&P and spackle both generates the same value compared with other literature. Fig. 10 shows the graphical comparison of NC against different attacks on Lena image.

Different attack results on pepper and baboon images in Tables 8 and 9 shows that the proposed method achieved results for all attacks as compared with [5] and slightly different from [35]. Moreover, Figs. 11 and 12 shows the graphical comparison for pepper and baboon image. The performance of attacks is very good instead of some cases where it shows deficiency in performance.

In general, the proposed method demonstrates strong performance across different attacks, maintaining high NC values and effectively conserve image content.

Moreover, the proposed technique is more robust with maximum average imperceptibility PSNR value 53.15 db, high payload and low computationally complexity in comparison with existing technique. The Comparison of robust nature of proposed technique with existing robust watermarking literature is shown in Table 10.

In depth description of robustness for different images given in Table 11. In which numerous attacks i.e., speckle, Gaussian, salt and pepper noise and cropping attacks are preformed on different images, to check the robustness of proposed technique based on NC and BER. Results in the given table shows that technique is defending against most of the attacks and validate the effeteness of the scheme.

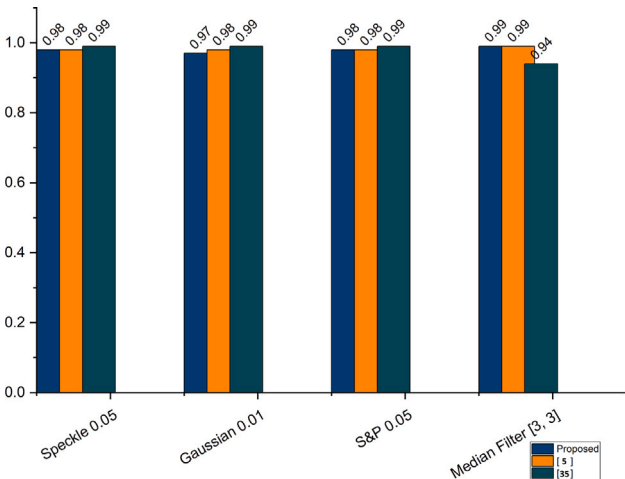


Fig. 12. Graphical comparison of NC for various attacks on the image baboon.

6. Conclusion

The security of multimedia data has become a challenge in today's digital landscape. The advancement in internet technologies has become a threat to digital data; now it has become easier to replicate, manipulate, and regenerate the digital data from internet sources, which undermines the security and integrity of the original content. This research explores the application of watermarking techniques in various domains, i.e., content verification, copyright protection, and tamper detection. In this paper, an image watermarking approach based on HED, LSB, and chaotic S-Box is proposed. A deep learning inspired HED model is used to identify sharp areas in the image. This step supports embedding watermark bits in the most suitable positions and making it difficult for the invader to estimate the watermark. To further enhance the security of the watermark signal, a chaotic S-Box is used to encrypt the watermark signal. The simulation results of the proposed approach show better visual quality and resistance against image processing and geometrical attacks. In most scenarios, the NC values achieved near to one mean the watermark was successfully recovered. The robustness parameter is just comparable for some scenarios, while for most cases, robustness comparison is better than state-of-the-art techniques. In the future, the robustness against various attacks can be further improved by refining the watermark embedding method. Additionally, research on video watermarking, where compression is still a major challenge, is possible.

CRedit authorship contribution statement

Muhammad Zubair Shoukat: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhaofeng Su:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Formal analysis, Data curation, Conceptualization. **Jehad Ali:** Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 10
Comparison of robust nature of proposed method with existing watermarking methods.

Technique	Purpose	Payload	Imperceptibility	Blind nature	Complexity
[2]	Data Hiding, Restoration, Authentication	High	45.39 dB	Blind	Low
[36]	Data Hiding, Security	High	51.5157	Non-Blind	Low
[37]	Data Hiding, Restoration, Authentication	Low	41.48 dB	Blind	High
[38]	Data Hiding, Restoration, Authentication	High	49 dB	Blind	Slightly higher
Proposed	Data Hiding, Security	High	53.15 dB	Semi-Blind	Low

Table 11
BER for different signal processing attacks.

Attacks	Lena				Baboon				Pepper				Plane				Sailboat			
	[8]		Proposed		[8]		Proposed		[8]		Proposed		[8]		Proposed		[8]		Proposed	
	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC	BER	NC
Gaussian noise (0.01)	0.09	0.962	0.017	0.966	0.022	0.956	0.019	0.962	0.026	0.946	0.019	0.962	0.021	0.958	0.022	0.956	0.026	0.946	0.03	0.940
Gaussian noise (0.02)	0.113	0.770	0.1005	0.799	0.121	0.753	0.1125	0.775	0.112	0.771	0.113	0.774	0.109	0.777	0.1185	0.763	0.099	0.799	0.127	0.746
Gaussian noise (0.05)	0.278	0.438	0.256	0.488	0.295	0.408	0.26	0.480	0.257	0.480	0.26	0.480	0.273	0.446	0.267	0.466	0.254	0.487	0.267	0.466
Speckle noise (0.01)	0.001	0.998	0.001	0.998	0	1	0.001	0.998	0	1	0.002	0.996	0.003	0.994	0.001	0.994	0.001	0.998	0.002	0.996
Speckle noise (0.02)	0.001	0.984	0.004	0.992	0.008	0.984	0.007	0.986	0.021	0.958	0.008	0.984	0.038	0.922	0.016	0.968	0.024	0.950	0.008	0.984
Speckle noise (0.05)	0.067	0.863	0.0095	0.981	0.074	0.849	0.02	0.960	0.091	0.816	0.01	0.980	0.173	0.647	0.018	0.964	0.088	0.820	0.024	0.952
S & P (0.01)	0.003	0.994	0.001	0.998	0.001	0.998	0.001	0.998	0.005	0.990	0.001	0.998	0.005	0.990	0.003	0.994	0.006	0.988	0.005	0.990
S & P (0.02)	0.027	0.946	0.004	0.992	0.012	0.976	0.001	0.988	0.028	0.942	0.007	0.986	0.023	0.952	0.01	0.980	0.024	0.950	0.008	0.984
S & P (0.05)	0.104	0.789	0.0075	0.985	0.088	0.822	0.009	0.982	0.104	0.788	0.01	0.980	0.109	0.778	0.019	0.962	0.110	0.778	0.02	0.960
Median filter [2, 2]	0.025	0.948	0.009	0.982	0.162	0.671	0.012	0.976	0.032	0.934	0.011	0.978	0.059	0.880	0.02	0.960	0.078	0.841	0.023	0.954
Median filter [3, 3]	0.011	0.978	0.004	0.992	0.138	0.719	0.005	0.990	0.014	0.972	0.005	0.990	0.026	0.946	0.009	0.982	0.050	0.899	0.014	0.972
Gaussian filter [2, 2]	0.020	0.960	0.009	0.982	0.151	0.692	0.0175	0.965	0.033	0.932	0.019	0.962	0.062	0.874	0.027	0.946	0.077	0.843	0.007	0.9849
Gaussian filter [3, 3]	0	1	0	1	0.002	0.996	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Crop 5%	0.004	0.992	0	1	0.004	0.992	0	1	0.004	0.992	0	1	0.005	0.990	0	1	0.005	0.990	0	1
Crop 10%	0.008	0.984	0	1	0.008	0.986	0	1	0.008	0.984	0	1	0.008	0.984	0	1	0.008	0.984	0	1

Data availability

Data will be made available on request.

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Muhammad Zubair Shoukat received the B.S. degree in Computer Science from COMSATS Institute of Information Technology Sahiwal, Pakistan, in 2014, and the MS degree from University of Lahore, Lahore Pakistan in 2018. He was with Government College University Faisalabad Sahiwal Campus as Lecturer from 2016 to 2023 and also with Bilytica Private Limited, Lahore Pakistan as web developer in 2015. His research interests include cryptography, privacy protection, deep learning and formal verification of distributed systems. He is currently pursuing his PhD degree in the School of Computer Science, University of Science and Technology of China.

Su Zhaofeng, is graduated from Wuhan University with a bachelor's degree in software engineering in 2012, and received his doctorate from the Quantum Software and Information Research Center of the University of Technology, Sydney in 2018. Joined the School of Computer Science, University of Science and Technology of China in January 2019. He has received the China Scholarship Council's National Study Abroad Scholarship for High-level Universities and the University of Technology Sydney's International Research Scholarship.

Dr. Jehad Ali (Senior Member IEEE, Top 2 Percent Scientist Globally according to Stanford University List Published by Elsevier in 2025) is currently working as an Assistant Professor at Department of AI Convergence Network, Ajou University, South Korea. Dr. Ali has a PhD in computer engineering from Ajou University, South Korea. Mr. Ali has teaching experience to graduate MS/PhD international students and undergraduate Computer Science/Engineering students. Besides teaching, Dr. Ali is actively performing research and has published more than 100 peer-reviewed papers in various journals and conferences such as IEEE Transactions on Industrial Informatics, IEEE Transactions on Intelligent Transportation Systems, IEEE Internet of Things Journal, IEEE Internet-of-Things magazine, Globecom Conference, IEEE Access, Sensors, Electronics, Applied Sciences, PloS One, etc., with a cumulative impact factor of more than 150 according to JCR 2021. Dr. Ali has a patent and has published three book chapters as a leading author. Mr. Ali has participated in and presented in IEEE and other prestigious conferences such as MobiSys (ACM), IEEE ICNC, IEEE IEMCON indexed by Scopus, DBLP, and Google Scholar and was held in the University of British Columbia Canada, Spain, and the United States of America. Dr. Ali is working as an Academic Editor for SCIE journal, such as IEEE Internet of Things, Springer Scientific Reports, and Telecommunication Systems Springer Nature. Moreover, Mr. Ali has worked as a Guest Editor of several special issues in different SCIE journals such as IEEE JBHI, Computers, Materials & Continua, Sensors, Intelligent Automation and Soft Computing, and Electronics. Mr. Ali has been peer reviewing papers for prestigious journals such as IEEE Transactions on Industrial informatics, Springer Wireless personal communication journal the Netherlands, IEEE ACCESS, IEEE Sensors, PloS One, Sensors, Electronics, Sustainability, Applied Sciences, Computer Networks (Elsevier), and IEEE communication Letters, etc. Dr. Ali has chaired sessions (Session Chair) in different International conferences such as ICUFN 2021, and ICNGC 2021 held in Jeju Island South Korea. A session chair in ICCCN 2022 (Which will be held in Manchester UK), ICACC E 2022 in Cyprus. In addition, Dr. Ali is in TPC of various IEEE conferences. Dr. Ali research interests include but are not limited to Software-Defined Networking, Extending it for 5G/6G issues, and improving QoS with SDN, Internet of Things, Wireless sensor networks, and applications of machine learning in Communication, Networking, and Healthcare.